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Math 208 Major Exam I Term 231

Original exam date: 2 October 2023. Practice version: questions and choices shuffled.
Net time allowed: 120 minutes

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1. Which one of the following statements is true about the subset V of $\mathbb{R}^3$ defined by

$$V = \{(x_1, x_2, x_3) : \ x_1 = 0\}?$$

(a) V is not closed under addition

(b) V is not closed under multiplication by scalars

(c) V is not closed under addition but closed under multiplication by scalars

(d) V is a subspace of $\mathbb{R}^3$

(e) V is closed under addition but not closed under multiplication by scalars

2. Find the order of the differential equation

$$\frac{d^3y}{dx^3} + \left(\frac{dy}{dx}\right)^8 + \sin^4 x = 0.$$

(a) 8

(b) 3

(c) 1

(d) 4

(e) 5

3. Find a constant A that makes the differential equation $(2x + Ay)\,dx + (3x + 2y)\,dy = 0$ exact.

(a) -2

(b) -3

(c) 2

(d) 3

(e) 0

4. Find a general solution of the differential equation $xy\dfrac{dy}{dx} = x^2 + 3y^2$.

(a) $2y^2 = cx^6 - x^2$

(b) $2y^2 = cx^5 - x^2$

(c) $2y^2 = cx^4 + x^3$

(d) $2y^2 = cx^5 + x^2$

(e) $2y^2 = cx^6 + x^2$

5. Find the sum of all values of r such that $y = e^{rx}$ is a solution of the differential equation $5y'' - 6y' - y = 0$.

(a) $-\dfrac{6}{5}$

(b) $\dfrac{3}{5}$

(c) $\dfrac{4}{5}$

(d) $-\dfrac{4}{5}$

(e) $\dfrac{6}{5}$

6. Find a general solution of the linear differential equation $x\dfrac{dy}{dx} = 2y + x^3\cos x$.

(a) $y(x) = x^3(\sin x + c)$

(b) $y(x) = x^2(\sin x + c)$

(c) $y(x) = x(\cos x + c)$

(d) $y(x) = x^2(\cos x + c)$

(e) $y(x) = x^3(\cos x + c)$

7. Find the value of k for which the vectors of $\mathbb{R}^3$

$$\mathbf{u} = (1, 1, 0),\ \mathbf{v} = (k, 3, 1),\ \mathbf{w} = (3, -2, -4)$$

are linearly dependent.

(a) $k = 0$

(b) $k = -\dfrac{3}{4}$

(c) $k = \dfrac{7}{4}$

(d) $k = \dfrac{3}{4}$

(e) $k = 2$

8. A particle is moving in a straight line with acceleration $a(t) = 18\cos(3t)$, and initial position $x(0) = -7$, and an initial velocity $v(0) = 4$. Find $x(\pi)$ (the position of the particle at $t = \pi$).

(a) $2\pi + 3$

(b) $4\pi - 3$

(c) $2\pi - 3$

(d) $4\pi + 3$

(e) $4\pi - 7$

9. In a certain culture of bacteria, the number of bacteria doubled after 3 hours. How long did it take for the population to triple?

(a) $\dfrac{\ln 16}{2}$

(b) $\dfrac{\ln 27}{\ln 2}$

(c) $\dfrac{\ln 3}{\ln 2}$

(d) $\dfrac{\ln 9}{\ln 2}$

(e) $\dfrac{\ln 12}{\ln 2}$

10. Find a general solution of the exact differential equation $(1 + ye^{xy})\,dx + (2y + xe^{xy})\,dy = 0$.

(a) $x + e^{xy} + y^2 = c$

(b) $x^2 + e^{xy} + y^2 = c$

(c) $x - e^{xy} + y^2 = c$

(d) $x + e^{xy} - y^2 = c$

(e) $x^2 - e^{xy} + y^2 = c$

11. Find a general solution of the differential equation $y' = \sqrt{x + y + 1}$.

(a) $3\sqrt{x + y + 1} + 2\ln\left(1 + \sqrt{x + y + 1}\right) = x + c$

(b) $3\sqrt{x + y + 1} + 2\ln\left(1 + \sqrt{x + y + 1}\right) = -x + c$

(c) $2\sqrt{x + y + 1} + 2\ln\left(1 + \sqrt{x + y + 1}\right) = -x + c$

(d) $2\sqrt{x + y + 1} - 2\ln\left(1 + \sqrt{x + y + 1}\right) = x + c$

(e) $3\sqrt{x + y + 1} - 2\ln\left(1 + \sqrt{x + y + 1}\right) = x + c$

12. Let $\mathbf{u} = (5, 1)$, $\mathbf{v} = (-2, -1)$, $\mathbf{w} = (5, -2)$ be vectors in $\mathbb{R}^2$. If $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$, then $a^2 + b^2 =$

(a) 8

(b) 16

(c) 25

(d) 36

(e) 34

13. Find the general solution of the differential equation $\dfrac{dy}{dx} = xe^{-x}$.

(a) $y(x) = 2(x + 1)e^{-x} + c$

(b) $y(x) = (x + 2)e^{-x} + c$

(c) $y(x) = -xe^{-x} + x + c$

(d) $y(x) = -(x + 1)e^{-x} + c$

(e) $y(x) = xe^{-x} + c$

14. Find a general solution of the separable differential equation $x^2\dfrac{dy}{dx} = 1 - x^2 + y^2 - x^2y^2$.

(a) $y(x) = \tan\left(c + \dfrac{1}{x} + x\right)$

(b) $y(x) = \tan\left(c - \dfrac{3}{x} + x\right)$

(c) $y(x) = \tan\left(c + \dfrac{2}{x} + x\right)$

(d) $y(x) = \tan\left(c - \dfrac{1}{x} - x\right)$

(e) $y(x) = \tan\left(c - \dfrac{2}{x} + x\right)$

15. The solution of the system

$$\begin{aligned} x_1 - 3x_2 - 7x_3 + 5x_4 &= 0 \\ 3x_1 + x_2 + 9x_3 - 5x_4 &= 0 \\ x_1 - 2x_2 - 4x_3 + 4x_4 &= 0 \end{aligned}$$

is the set of all scalar multiples of a vector $\mathbf{u}$ where $\mathbf{u} =$

(a) $(-2, -3, 1, 0)$

(b) $(-2, 3, 0, 1)$

(c) $(-1, -6, 1, 0)$

(d) $(-1, 3, 1, 0)$

(e) $(-2, 3, 1, 2)$

16. If $\mathbf{u}, \mathbf{v}$ and $\mathbf{w}$ are vectors in $\mathbb{R}^3$ and r and s are real numbers, then which one of the following statements is not true?

(a) $\mathbf{u} \cdot \mathbf{u} = |\mathbf{u}|$

(b) $r(\mathbf{u} + \mathbf{v}) = r\mathbf{u} + r\mathbf{v}$

(c) $(\mathbf{u} + \mathbf{v}) + \mathbf{w} = \mathbf{u} + (\mathbf{v} + \mathbf{w})$

(d) $(r + s)\mathbf{u} = r\mathbf{u} + s\mathbf{u}$

(e) $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$

17. Solve the initial-value problem $y' + y = 2$, $y(0) = 0$.

(a) $y(x) = 1 + e^{-x}$

(b) $y(x) = 2 - 2e^{-x}$

(c) $y(x) = 3 - 2e^{-x}$

(d) $y(x) = 2 - e^{-x}$

(e) $y(x) = 1 - e^{-x}$

18. Find a value of c such that $y = \tan(x^3 + c)$ is a solution of the initial value problem $y' = 3x^2(y^2 + 1)$, $y(0) = 1$.

(a) $\dfrac{3\pi}{4}$

(b) $\dfrac{5\pi}{3}$

(c) $\dfrac{\pi}{3}$

(d) $\dfrac{5\pi}{4}$

(e) $-\dfrac{\pi}{4}$

19. Find a general solution of the differential equation $y'' = 2y(y')^3$.

(a) $y^3 + 3x + Ay + B = 0$

(b) $y^2 + 3x + Ay + B = 0$

(c) $y^3 - 3x + Ay + B = 0$

(d) $y^3 + 2x + Ay + B = 0$

20. By making a suitable substitution, the differential equation $x^2y' + 2xy = 5y^3$ can be transformed into a linear differential equation

(a) $v' + \dfrac{3}{x}v = \dfrac{5}{x^2}$

(b) $v' - \dfrac{4}{x}v = -\dfrac{10}{x^2}$

(c) $v' - \dfrac{4}{x}v = \dfrac{5}{x^2}$

(d) $v' - \dfrac{5}{x}v = \dfrac{10}{x^2}$

(e) $v' - \dfrac{3}{x}v = -\dfrac{5}{x^2}$

Answer Key

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Q	Ans	Brief explanation
1	**(d)**	$x_1 = 0$ is closed under addition and scalar multiples, so V is a subspace.
2	**(b)**	Highest derivative is y''', so the order is 3.
3	**(d)**	$M_y = A$, $N_x = 3$, exactness gives $A = 3$.
4	**(a)**	$v = y^2$: $v' - \frac{6}{x}v = 2x$, $\rho = x^{-6}$, so $2y^2 = cx^6 - x^2$.
5	**(e)**	$5r^2 - 6r - 1 = 0$, roots $\frac{3\pm\sqrt{14}}{5}$, sum $= \frac{6}{5}$.
6	**(b)**	$\rho = x^{-2}$: $(x^{-2}y)' = \cos x$, so $y = x^2(\sin x + c)$.
7	**(c)**	$\det = 4k - 7 = 0$, so $k = \frac{7}{4}$.
8	**(b)**	$v = 6\sin 3t + 4$, $x = -2\cos 3t + 4t - 5$, so $x(\pi) = 4\pi - 3$.
9	**(b)**	$e^{3k} = 2$, $t = \frac{\ln 3}{k} = \frac{3\ln 3}{\ln 2} = \frac{\ln 27}{\ln 2}$.
10	**(a)**	$F = x + e^{xy} + g(y)$, $g' = 2y$: $x + e^{xy} + y^2 = c$.
11	**(d)**	$v = x + y + 1$, $v' = 1 + \sqrt{v}$, $u = \sqrt{v}$: $2u - 2\ln(1+u) = x + c$.
12	**(e)**	$5a - 2b = 5$, $a - b = -2$: $a = 3$, $b = 5$, so $a^2 + b^2 = 34$.
13	**(d)**	By parts: $\int xe^{-x}dx = -xe^{-x} - e^{-x} + c = -(x+1)e^{-x} + c$.
14	**(d)**	$(1-x^2)(1+y^2)$: $\arctan y = -\frac{1}{x} - x + c$.
15	**(a)**	Row reduce: $x_4 = 0$, $x_2 = -3x_3$, $x_1 = -2x_3$, so $\mathbf{u} = (-2, -3, 1, 0)$.
16	**(a)**	$\mathbf{u}\cdot\mathbf{u} = \lvert\mathbf{u}\rvert^2$, so $\mathbf{u}\cdot\mathbf{u} = \lvert\mathbf{u}\rvert$ is false.
17	**(b)**	$\rho = e^x$: $(e^x y)' = 2e^x$, $y = 2 + Ce^{-x}$, $C = -2$.
18	**(d)**	$\tan c = 1 \Rightarrow c = \frac{\pi}{4} + k\pi$; $k = 1$ gives $c = \frac{5\pi}{4}$.
19	**(a)**	$p = y'$: $\frac{dp}{p^2} = 2y\,dy$, $(y^2 + C_1)dy = -dx$: $y^3 + 3x + Ay + B = 0$.
20	**(b)**	Bernoulli $n = 3$, $v = y^{-2}$: $v' - \frac{4}{x}v = -\frac{10}{x^2}$.